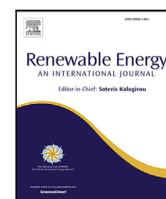




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Performance evaluation of vertical bifacial and single-axis tracked agrivoltaic systems on arable land

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ABSTRACT

This study evaluates and compares two types of interspersed bifacial agrivoltaic systems in Belgium: a fixed vertical system and a dynamic single-axis tracker, focusing on sugarbeet cultivation. Additionally, the impact of different tracking algorithms on crop yield and quality is investigated. The main approach involves developing an empirical crop model based on radiation use efficiency (RUE) and utilizing a radiation tool to simulate crop yield and quality. Field measurements of electricity and crop output are conducted over a span of two years, 2021 and 2022. The findings reveal that the dynamic solar tracker outperforms the fixed vertical bifacial setup in both years. The smart-tracking algorithm, applied under optimal watering conditions in the 2021 season, leads to a significant increase in energy yield (+30%) and land use efficiency (+20%) at a lower cost, while maintaining comparable crop yields to the vertical setup. However, limitations are observed in the empirical crop growth model's ability to account for climatic variability in the dry 2022 year, reducing its usefulness during the design phase. Notably, the practical implementation of the systems highlights challenges during construction and exploitation, emphasizing the significance of considering these practical factors together with climate, structure and crop choice assessing the effectiveness and to de-risk investments.

1. Introduction

Electricity and heat production are major contributors to greenhouse gas emissions, leading to environmental issues associated with climate change [1]. To address these challenges sustainably, there is a need to increase the share of renewable energy sources. Solar photovoltaics (PV) is one of the most prominent renewable energy sources that can contribute to meeting the growing energy demand [2]. Currently, PV modules are mainly deployed on rooftops and marginal land areas [3]. However, to accelerate the widespread adoption of PV and meet energy demands sustainably, exploring alternative applications beyond traditional settings becomes crucial.

In recent years, the exploration of innovative approaches to harness solar energy has gained considerable attention in the field of renewable energy research. Building-integrated photovoltaics (PV), floating solar farms, solar carports, and other novel concepts have been extensively studied as alternatives to conventional ground-mounted PV systems. Among these emerging alternatives, agrivoltaics has emerged as a promising and compelling concept with significant theoretical potential. Agrivoltaics represents an integration of agricultural production and solar electricity generation on the same land, enabling a dual-purpose utilization of land resources [4,5].

The concept of agrivoltaics was first proposed by Goetzberger and Zastrow in 1981, recognizing the possibility of combining agricultural activities with solar energy production. Subsequently, Nagashima further advanced the concept by developing and implementing several prototypes of agrivoltaic systems in Japan [6,7]. The underlying idea behind agrivoltaics is to maximize the productive use of land by concurrently generating renewable energy and cultivating crops. This integration presents a unique opportunity to enhance land utilization efficiency and promote sustainable development by addressing the challenges of limited land availability and increasing demands for both food and renewable energy [8].

Agrivoltaic systems can be broadly categorized into two main groups: overhead agrivoltaics and interspersed agrivoltaics [9,10]. Overhead agrivoltaics involve installing PV modules at a certain height to allow machinery to operate underneath them. On the other hand, interspersed agrivoltaics place PV modules closer to the ground, requiring machinery to drive between the rows. Both configurations can utilize fixed or dynamic (tracking) PV installations.

To date, the majority of research studies in the field of agrivoltaics have primarily focused on overhead systems, whether fixed or dynamic in nature. For instance, Trommsdorff et al. [11] conducted a study where PV modules were fixed at a height of 5 meters and oriented

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southwest to ensure a uniform distribution of solar radiation. In France, Marrou et al. [12] investigated the impact of agrivoltaics on lettuce, cucumber, and wheat by arranging PV arrays at full and half densities with row-to-row spacing at a height of 4 m. Another study in France by Valle et al. [13] explored dynamic systems using lettuce and introduced an innovative tracking strategy involving anti-tracking in the morning, which led to improved land use efficiency.

In Italy, Amaducci et al. [14] conducted simulations to evaluate an innovative algorithm's ability to maintain adequate crop yield levels in agrivoltaic systems. However, when it comes to interspersed agrivoltaic systems, the research is relatively limited and focused on fixed vertical bifacial structures and traditional north–south layouts. Riaz et al. [15] investigated the potential of vertical facing bifacial PV farms for agrivoltaic systems, specifically analyzing the distribution and amount of photosynthetically active radiation and energy yield. Their results indicated that vertical PV designs enhance spatial homogeneity compared to traditional north–south tilted PV layouts. Similarly, Campana et al. [16] studied the optimization of agrivoltaic systems using a theoretical crop model, albeit not validated under shaded conditions. Their findings emphasized the need to consider multiple performance indicators during system optimization, with the row distance being identified as the main variable influencing both crop yield and energy yield.

Although research on overhead agrivoltaic systems has provided valuable insights, further investigation into interspersed systems is warranted to explore their potential and understand their performance under different conditions. There are several research gaps when it comes to the practical and theoretical implementation of interspersed agrivoltaic (agriPV) systems. These gaps highlight areas where further investigation and understanding are needed for successful integration of agriPV systems with existing agricultural practices.

Firstly, there is a limited amount of research focusing specifically on a theoretical framework for the structural design and control algorithm with unique challenges due to the complex interaction between the crops and the PV modules, as well as the dependence on external factors such as climate and environmental conditions. In this study, an empirical crop model, based on real-world observations and measurements, has been implemented to offer a practical approach to estimate crop yield and assess the potential benefits of agrivoltaic systems. The advantage of this study is the validation of the effectiveness of the empirical crop model under various climatic and structural scenarios based on 2 year data.

Another significant research gap pertains to the structural design and algorithm (solar tracker vs vertical fixed set-up) and impact on crop yields, energy production, cost and practical land loss (and final investment decision). Additionally, optimizing the structural design (vertical bifacial or solar tracking) of the agriPV system to minimize crop yield losses is critical research gap that needs to be addressed. To do so, a practical comparison is conducted between two popular interspersed agrivoltaic systems combined with arable field crops, located in Grembergen, Belgium. The study examines a fixed vertical bifacial system and a dynamic single-axis tracker, considering various performance indicators such as crop yield, quality, energy yield, cost, and spatial efficiency. The research involves theoretical modeling as well as field measurements gathered during two growing seasons (2021 and 2022), with sugar beets (*var. KWS Laurena*) as the cultivated crop. Different tracking strategies, including smart-tracking during rainy seasons and standard solar tracking during hot and dry periods, are evaluated to provide comprehensive insights into the decision-making process for interspersed agrivoltaic systems.

Finally, this paper's ultimate strength lies in its practical implementation approach, which not only fosters comprehension but also sheds light on challenges inherent in system construction and operation. The substantial land loss incurred during practical cultivation is meticulously quantified, underscoring its influence on spatial efficiency. Additionally, the ramifications of soil compaction, weed proliferation,



Fig. 1. Overview of the Grembergen agrivoltaic site, showing the fixed vertical bifacial set-up and dynamic single-axis bifacial tracker.

agricultural yield variability throughout the years and photovoltaic soiling effects are examined. These critical variables are then discussed within the framework of the purchase power agreement business model, recognizing their substantial impact on the entire system's viability.

2. Materials and methods

To ensure experimental control and minimize the influence of external variables such as soil variability, wind speed, and air temperature, both the fixed vertical and single-axis tracking agrivoltaic systems were installed on the same field in Grembergen, Belgium (51.02°N, 4.12°E). The row-to-row distance of 9 m was carefully selected to accommodate farming practices and the required machinery while maintaining appropriate spacing for crop growth and accessibility. This decision ensures that the agrivoltaic system can be seamlessly integrated into existing agricultural operations without significant disruptions or limitations (8 m width plus 0.5 m spare on each side). The orientation of the structures follows the field direction (25° southeast) to optimize the total cultivation area. Considering the nearby residential area, the maximum height of 2.6 m was determined to minimize any visual impact. By maintaining a low-profile configuration, the agrivoltaic systems blend harmoniously with the surrounding landscape, preserving the aesthetic appeal of the area. To minimize visual impact on the nearby residential area, the maximum height of the structures was set at 2.6 m. Consequently, both setups were limited to a 2 pair module landscape (2L) configuration. Additionally, the electrical grid capacity played a role in determining the installation size. With a maximum additional injection limit of 24 kVA, the number of modules in each setup was restricted to 72, with each module having a power rating of 455 Wp. This constraint ensures compliance with grid regulations and limitations without incurring excessive costs associated with grid reinforcement.

2.1. Practical implementation

As depicted in Fig. 1, the first agrivoltaic setup utilizes a fixed configuration. It comprises three rows of vertically installed bifacial PV modules (Phono solar Half-cut PERC 455 Wp) with dimensions of 2.13×1 m and a bifaciality factor of 0.7. This arrangement results in a power density of 450 kWp/ha. Each row consists of 12 modules, with two PV modules stacked in a landscape position, installed 0.6 m above the ground to prevent shading of the crops. A clearance of



Fig. 2. The bifacial single-axis solar tracker optimized to increase the bifacial gain. There is an increased gap between the torque tube and modules, limiting structural shading on the backside of the module.

0.15 m is intentionally maintained between the south-facing vertical pillars and the PV modules to avoid shading by the pillars and reduce lateral wind load. The front side of the PV modules faces the west side to maximize energy yield, considering the southeast orientation. Each row is connected as a unique string, feeding into the inverter's Maximum Power Point (MPP) input. Due to limited grid capacity, the SMA Tripower 6.0 kVA inverter was selected, resulting in a DC/AC ratio of 1.82. This choice facilitates easy measurement of (string) voltage and current.

The second agrivoltaic setup features a dynamic bifacial single-axis tracking system with a similar module layout and power capacity as the vertical setup. The single-axis tracker structure is designed to optimize energy yield compared to a standard monofacial tracker. To enhance bifacial gain, the structural elements at the backside are minimized by increasing the gap between the torque tube and module (50 cm) and ensuring longitudinal module spacing (1 cm), as illustrated in Fig. 2. With a height of 2.3 m, the single-axis tracker intercepts as much light as possible. Linear actuators control the tracker, and its position is determined based on solar irradiance (diffuse mode), wind speed, sun position, and the chosen tracking algorithm, which is implemented in an industrial Programmable Logic Controller (PLC). The tracking angles range from -50° (east) to $+50^\circ$ (west), with 0° representing a horizontal position. This system has the ability to enhance power production by orienting itself towards the sun and is connected to the SMA Tripower 8.0 kVA inverter (DC/AC ratio 1.36) to accommodate its increased output.

Both agrivoltaic setups are installed using the pile drilling technique, as illustrated in Fig. 3. This method is commonly employed in ground-mounted PV system foundations and offers several advantages over concrete foundations. Pile drilling is a cost-effective and time-efficient process, and it is reversible, allowing for quick and easy removal without causing permanent soil damage. The depth of the foundation is determined based on soil type profiles obtained through pull-out tests and the wind-load requirements, typically ranging between 2 and 3.8 m. The steel framework for both the vertical fixed and single-axis tracker is attached to these vertical pillars. Additionally, as shown in Fig. 4, the bifacial PV modules are mounted in a landscape position. Subsequent steps involve the installation and calibration of the linear actuators and motor controllers for the solar tracker. Once these components are in place, the Programmable Logic Controller (PLC) system is installed and programmed to optimize the operation of the solar tracker.

To prevent any interference during field cultivation, all underground DC and data cabling is buried at a depth of 1 meter. Since this farm focuses on arable farming, there is no significant on-site electrical

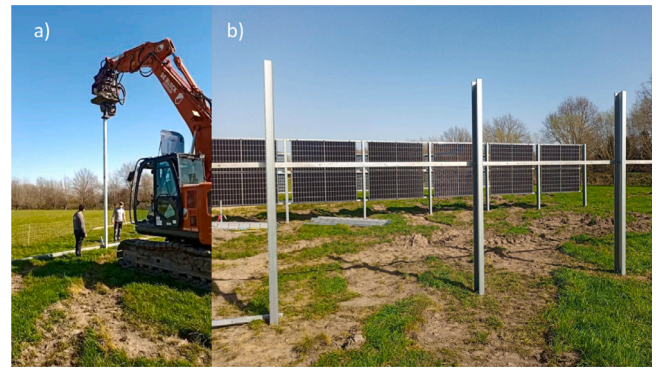


Fig. 3. Building process. (a) Pile drilling of the vertical profiles (b) construction of the remaining steel framework where the PV modules can be attached.

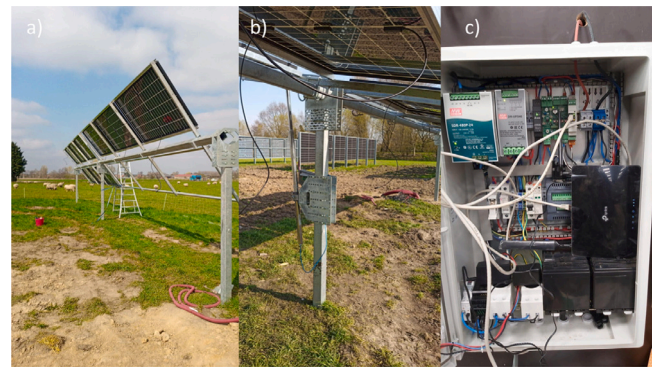


Fig. 4. Last step in the building process: the PV modules are placed, linear actuators for the solar tracker are placed and the PLC system is connected and programmed.

consumption, such as cooling systems. As a result, the agrivoltaic installation functions as a net-producing system, where the surplus energy generated is sold back to the grid.

The technical and environmental aspects of the project were consolidated into a building permit application, which was submitted in March 2020 and subsequently approved by local authorities in August 2020, following advice from the Ministry of Agriculture and Fisheries. Construction of the agrivoltaic installation took place in March 2021, and after undergoing an electrical inspection and receiving approval from the distribution system operator (DSO), the system was fully operational by July 2021.

2.2. Measurement system

Starting from July 2021, the agrivoltaic systems have been equipped with a data acquisition system based on MODBUS RTU. This system allows for the connection of up to 247 sensors, reducing the need for extensive cabling (only four wires - GND, 24 V, DATA+, DATA-) and enjoys broad compatibility with various sensor brands and types. Fig. 5 illustrates the placement of agri-, PV-, and meteorology-related sensors throughout the agrivoltaic systems:

- Agri micro-climate: Apogee SQ442-SS PAR sensors
- PV: IMT solar Si reference cells, PT1000 module sensors, and inverter data
- Meteorology: PT1000 temperature sensor and anemometer

As depicted in Fig. 5, three PAR sensors were positioned between two rows of the vertical bifacial setup to evaluate the cumulative PAR reaching the crops. Both the vertical and tracking setups feature plane of array (POA) Si-reference cells and module temperature sensors

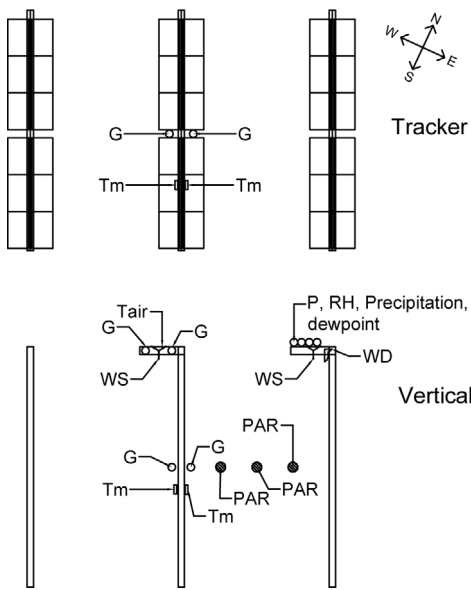


Fig. 5. Schematic overview of the installed agrivoltaic sensors at the Grembergen setup: reference cells (G), PAR sensors (PAR), temperature sensors (Tm - module, Tair - ambient), wind speed (WS) and direction (WD).

(front- and backside) to compare the expected energy yield with the actual yield obtained from the inverters. Additionally, a global horizontal irradiance (GHI) Si-reference sensor and a reversed GHI sensor (facing the ground) were installed to measure GHI and estimate the crop albedo. Considering the spectral effects on bifacial PV performance estimation, no thermopile pyranometer was implemented based on the findings of Gostein et al. [17], which highlighted a significant spectrally responsive albedo mismatch ($\pm 9.2\%$) compared to reference cells.

To assess the microclimate, ambient air temperature and wind speed sensors were installed. All these sensors were logged at a resolution of 5 min and the data was stored in a cloud-based database, ensuring efficient and centralized data management.

2.3. Theoretic crop framework

To better assess the impact of different tracking strategies on crop yield, empirical data from Artru et al. [18] is utilized to examine the development of sugar beets under dynamic, full, and no shade conditions in equivalent environmental circumstances (Belgium, optimal watering conditions). Their data reveals an increased radiation use efficiency (RUE) in mild shade levels, which can be attributed to morphological changes that counteract the reduced radiation (such as higher specific leaf area) and increased penetration of diffuse radiation deeper into the canopy. Fig. 6 presents the corresponding shade response curve (SRC) for both total beet (root) biomass and sugar mass for the 2015 and 2016 growing seasons. It can be observed that both total root biomass and sugar mass exhibit the same trend, indicating a consistent relative sugar content (%).

The quality of sugar beets can be evaluated by the total extractable sugar content. This value can be calculated by multiplying the total sugar content (Fig. 6) with the extractability factor (Fig. 7). The extractability of sugar beets is influenced by various factors, including environmental conditions and field management practices such as fertilization and irrigation, as these can affect the presence of other components like potassium and sodium. In line with the findings of the aforementioned study [18], Fig. 7 demonstrates a strong correlation between extractability and the total received global irradiance, indicating the significance of solar radiation in determining sugar beet extractability.

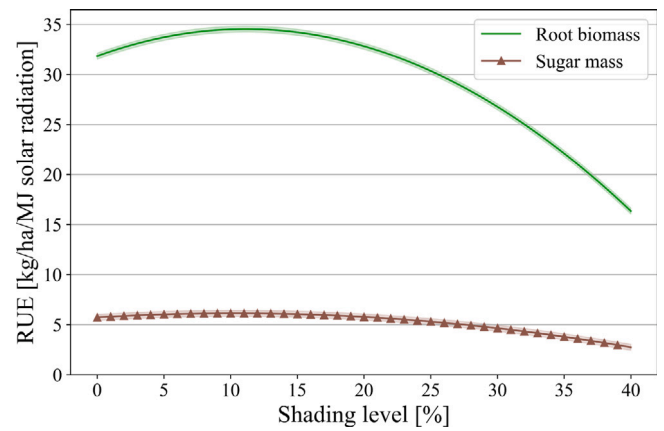


Fig. 6. Fitted shade response curve for sugar beet cultivation in Belgium, data from [18]. The radiation use efficiency (RUE) increases at low shading levels but decreases drastically at higher shade levels.

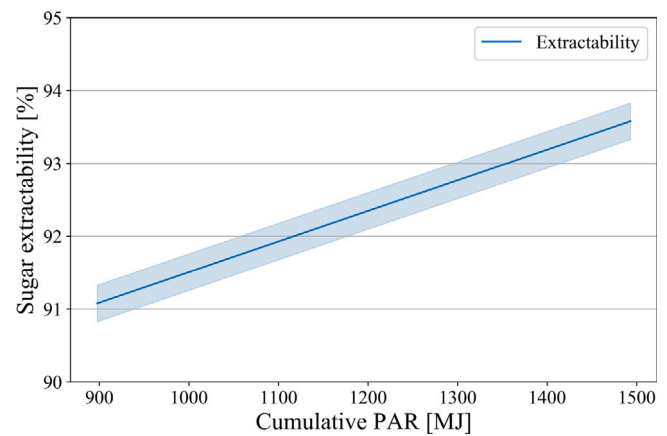


Fig. 7. The fitted sugar extractability of the sugarbeets with respect of the cumulative radiation level showing that more sugars are created when a higher amount of radiation is present.

2.4. Tracking algorithms

In contrast to the fixed vertical bifacial setup, which has fixed geometrical parameters due to design constraints, the dynamic single-axis tracker offers the ability to dynamically control shade levels for crops based on time and location. In this study, two binary positions of the tracker were defined: “tracking” and “anti-tracking”. During “tracking”, the PV modules are oriented perpendicular to the sun, maximizing energy production but increasing ground shade fraction. During “anti-tracking”, the modules are aligned with the sun, increasing the transmitted amount of light but reducing energy production potential. The timing and duration of “tracking” and “anti-tracking” can be adjusted on different temporal scales, ranging from minutes to months.

Two different tracking algorithms were discussed and compared: full tracking, which includes standard solar tracking and back-tracking at low sun angles to minimize self-shading, and smart tracking, which incorporates a combination of tracking and anti-tracking periods. Full tracking is applied during dry and hot summers to protect crops against water stress, while smart tracking is used in rainfed years to mitigate crop losses. The full tracking algorithm optimizes module position to maximize energy production and shade levels by intercepting direct normal irradiance (DNI) using an astronomical sun position algorithm [19], as well as diffuse horizontal irradiance (DHI) based

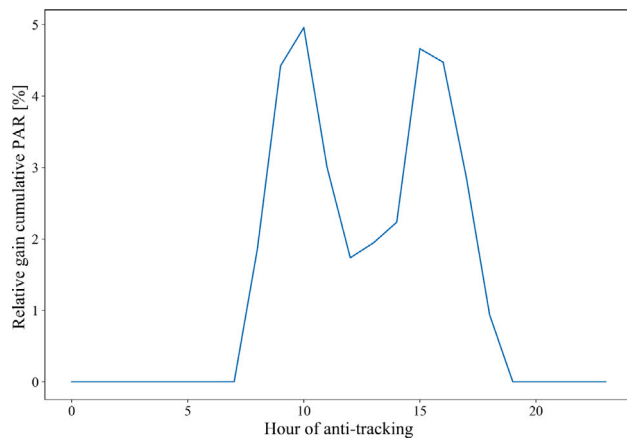


Fig. 8. Optimal moments to anti-track to increase the cumulative amount of PAR radiation in an interspersed east–west single-axis system. During morning and evening, PAR values are too small to have a large influence and shade fractions are out of interest during the midday.

on real-time measurements of global horizontal irradiance (GHI). The tracker rotates to a horizontal position if GHI falls below 200 W/m².

During smart tracking, a trade-off between tracking (energy yield) and anti-tracking (crop yield) is achieved based on daily photosynthetic and energy production.

In our interspersed east–west tracking system in Belgium, optimal moments for anti-tracking were defined to optimize the radiation use efficiency (RUE) of sugar beets (shade level around 10%) for all crops in the field (Fig. 6). Anti-tracking during periods of low PAR in the morning and evening has a limited impact on the cumulative radiation amount. At noon, when the sun is directly in front of the tracker, shade fractions below have no impact on the cultivated area (sugar beets). Consequently, the optimal moments with the greatest impact on cumulative radiation amount (shade fraction and PAR level) were identified between 10–11 am and 2–3 pm as shown in Fig. 8.

Using a ground radiation tool based on a view factor model developed in previous research [20], Fig. 9 illustrates the distribution of PAR radiation between two PV structures from April to October. In the fixed vertical setup, shade occurs in the early morning and late evening, resulting in the longest shading duration for crops closest to the PV modules [21], and an average PAR percentage of 79%. The full tracking setup exacerbates this effect, leading to very low relative PAR percentages near the structure and a large spread in radiation values, with an average PAR percentage of 75%. The smart-tracking algorithm with anti-tracking moments proves to be useful, as it increases relative PAR values near the structure, improving the homogeneity of PAR distribution and resulting in a similar average PAR percentage of 79% compared to the vertical setup. However, monthly differences in PAR distribution still occur due to variations in sun angles and subsequent shade fractions, as shown in Fig. 10. For illustrative purposes, Fig. 9d depicts the light distribution if anti-tracking were continuously active, considering the physical limitations of the tracking installation, where some light is still intercepted by the PV modules.

3. Results

In this paper, a comparison was made between the performance of the vertical bifacial set-up and the dynamic single-axis tracker. Throughout the rainy growing season of 2021 (spanning from April 2021 to October 2021), the smart tracking algorithm was employed. However, for the hot and dry season of 2022, characterized by limited rainfall, the decision was made to switch to traditional full (energy) tracking. The precipitation levels, average air temperature, and solar hours recorded during the two seasons are presented in Fig. 11. The

measured and theoretical results according to the crop, energy, financial and spatial yield are shown as well as important practical aspects during the construction and exploitation.

3.1. Electricity production

Fig. 12 presents the measured monthly AC electricity production (kWh) for each system, both operating at equal power density. The data clearly demonstrates that the dynamic single-axis tracker outperforms the fixed vertical setup, resulting in approximately a 35% increase in monthly electricity production. Notably, the difference in tracking strategy between 2021 and 2022 is evident. For instance, comparing the months of July, August, and September in 2021 (smart tracking) to the same period in 2022 (full tracking), the electricity production shows a significant disparity, with a +15% increase in 2021 and a more substantial +35% increase in 2022.

Over the course of the first year of operation, from July 2021 to July 2022, the specific yield for the dynamic single-axis tracker was 1305 kWh/kWp, while the fixed vertical bifacial setup achieved a specific yield of 915 kWh/kWp. These specific yield values provide insights into the energy production efficiency of each system, highlighting the superior performance of the dynamic single-axis tracker. As a reference typical South-facing ground mounted installation, with the same bifacial modules, a tilt of 20° and ground coverage ratio of 55%, the specific yield would be 1045 kWh/kWp. However, given the low ground coverage ratio of the agrivoltaic installations (resulting in 450kWp/ha) compared to the reference South-facing installation (1140kWp/ha), the electricity yield per ha is 60% lower for the vertical bifacial system and 45% lower for the single-axis tracker, both compared to the reference South-facing installation.

Fig. 13 illustrates the daily production profiles under different sky conditions. An initial observation is the similar start and end times of production for both the dynamic single-axis tracker and the fixed vertical setup, despite the mechanical limit of the tracker being 50° and the vertical setup being fixed at a 90° angle. On cloudy days, the tracker is capable of assuming a flat position, enhancing light interception and electrical production by approximately 50% compared to the vertical setup. In sunny conditions, the dynamic tracking system, which continuously optimizes the angle of incidence of sun rays, outperforms the vertical setup, particularly during midday when the difference is most pronounced. Fig. 13 also highlights the disparity between full energy tracking and smart tracking. During anti-tracking moments (indicated by the drops at 10 am and 18 pm), the electrical production diminishes to approximately 25% of the nominal power.

3.2. Crop development, yield and quality

Utilizing the theoretical empirical and a cumulative radiation of 2733 MJ during the 2021 growing season, Fig. 14 estimates a reduction of 26% for full solar tracking, 20% for the vertical setup, and 15% for the smart tracking configuration. Despite the vertical setup and smart tracking having a nearly equal average cumulative amount of PAR across the field, there is an increase in annual agricultural yield and quality when applying the smart tracking algorithm. This disparity can be attributed to spatial integration and the distribution of PAR. Although the average relative PAR amount remains the same, the smart tracking approach ensures that a larger number of plants receive a consistent high amount of cumulative radiation, resulting in higher RUE values. In contrast, the vertical setup experiences a wider range of PAR levels across the field, with some areas receiving significantly lower amounts of PAR. This discrepancy leads to reduced RUE and, consequently, lower crop yield in those areas.

Similar results were measured in the field. In the rain-fed growing season of 2021, the root biomass of sugar beets decreased by 26% in the vertical setup, whereas in the smart tracking setup, the reduction was only 22%. However, it should be noted that there was significant variability in crop yield, primarily due to soil conditions after installation,

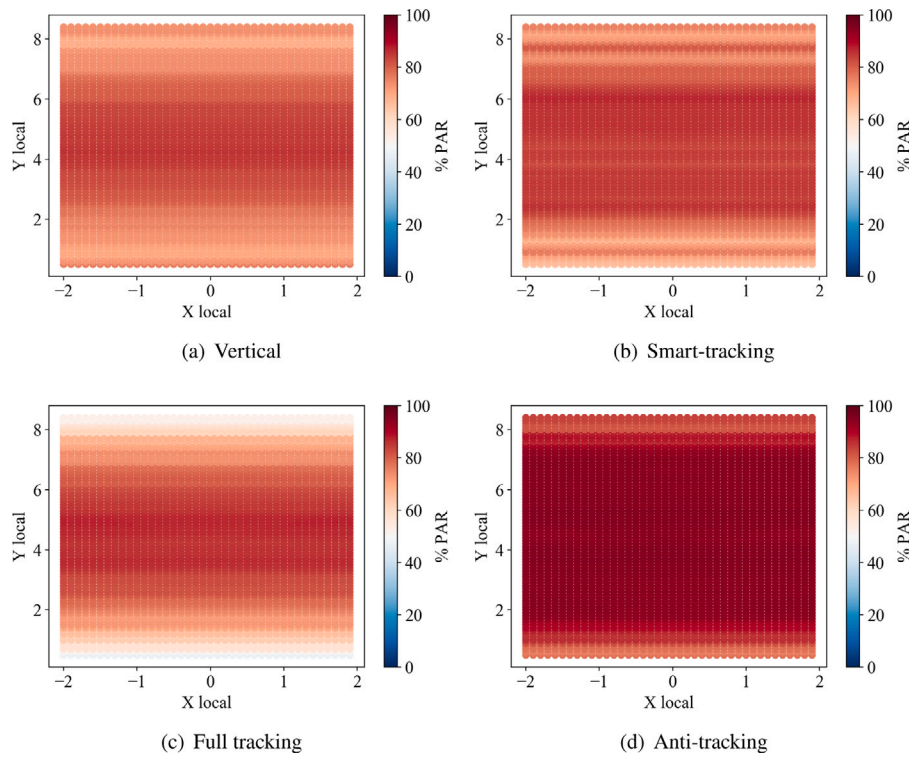


Fig. 9. The relative PAR distribution compared to open field conditions (100% PAR) between two interspaced structures showing that the largest PAR reductions are located close to the structure (with larger effects for tracking than the vertical system).

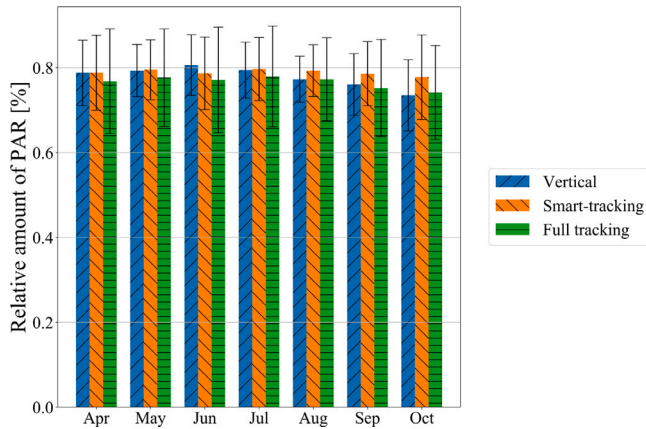


Fig. 10. (Spatial averaged) monthly remaining PAR percentages below the vertical and solar tracking set-up showing that almost 80% of the PAR light remains available for crop growth. Error bars represent the standard deviation of the PAR distribution.

including heavy compaction and distortion, making it challenging to draw definitive conclusions. Additionally, the total biomass reduction was observed to be 17% in the vertical setup and 12% in the smart tracking setup, indicating that a greater proportion of assimilates were allocated to stems and leaves, resulting in a decreased harvest index.

In the dry year of 2022, with limited rain events, the root biomass of sugar beets in the vertical setup was comparable to the reference area, while a loss of 10% was measured for the full tracking setup. In both years, quality remained equal between the agrivoltaic beets and reference beets.

3.3. Spatial efficiency

The dual land use of agrivoltaics can increase the spatial efficiency. A commonly used parameter in agrivoltaics to express the spatial

efficiency is the Land Equivalent Ratio (LER) which can be calculated according to Equation (1).

$$LER = \frac{Y_{agri,AV}}{Y_{agri,ref}} * (1 - LL) + \frac{Y_{el,AV}}{Y_{el,ref}} \quad (1)$$

The LER compares the electrical and agricultural yields ($Y_{el,AV}$ (MWh) and $Y_{agri,AV}$ (kg)) per unit of land (ha) between agrivoltaic and reference conditions. Consequently, the values of the LER are highly dependent on the reference situation. In this study, the reference ground-mounted PV is considered with the same type of bifacial modules, where the PV modules do not fully cover the agricultural area. The coverage ratio (55%), height (1m) and tilt (20°) are adapted to have at least 30% available sunlight, maintaining soil fertility and biodiversity. This adaptation is needed since the LER does not evaluate the environmental impact between a reference and agrivoltaics solar farm. The same reasoning is made for the agricultural yield. It is the marketable yield (kg/ha) that is compared with the reference (i.e. quality parameters such as sugar content, shape, color are included). Important is to reduce the agricultural area in agrivoltaic conditions by a landloss factor LL (between 0 and 1) due the presence of the agrivoltaic structure and borders.

Theoretical calculations initially accounted for a buffer zone of 0.5 m on both sides of the agrivoltaic structure, which would result in a land loss factor of 11% (1 m divided by 9 m row distance). However, as depicted in Fig. 15, in practice, the buffer zone was measured as 1 m on each side due to the absence of GPS-tracked machinery. Consequently, the practical land loss amounted to 22%.

As tabulated in Table 1, based on the significant land loss, reduction in root biomass and reduction in electricity yield per ha, the vertical system in the rain-fed 2021 season achieved a Land Equivalent Ratio (LER) of 0.95. This indicates that there was no improvement in spatial efficiency compared to separate production systems. In contrast, the smart tracking system achieved an LER of 1.15, primarily due to its higher specific energy yield when compared to the vertical system.

In the dry 2022 season, which allowed for full tracking capability resulting in a 35% increase in energy compared to the vertical setup,

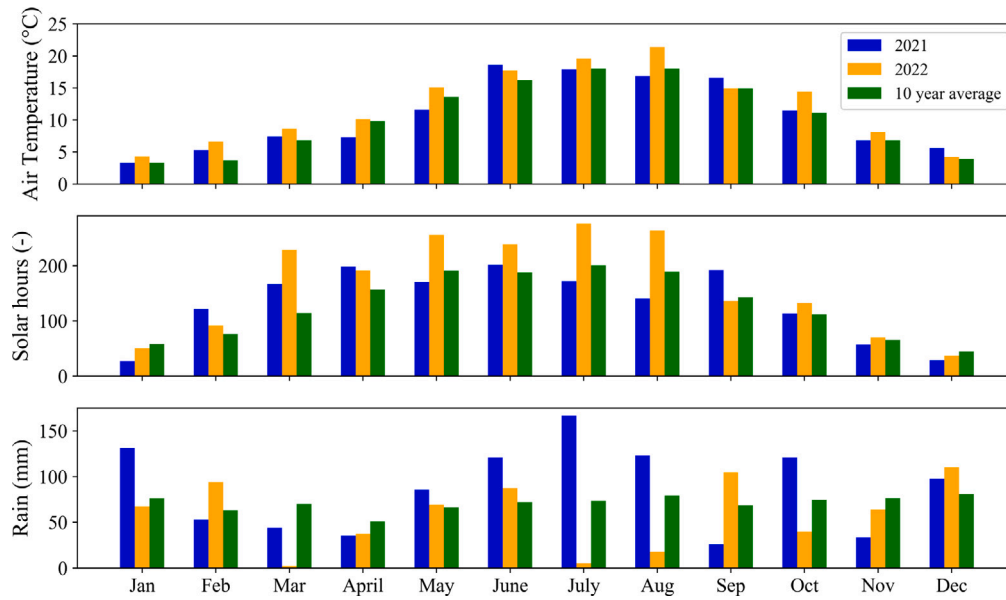


Fig. 11. The air temperature, amount of rain and solar hours in year 2021 and 2022, Belgium. This figure shows the significantly difference between two seasons: 2022 had a lot more sun hours, a higher temperature and almost no rain compared to 2021.

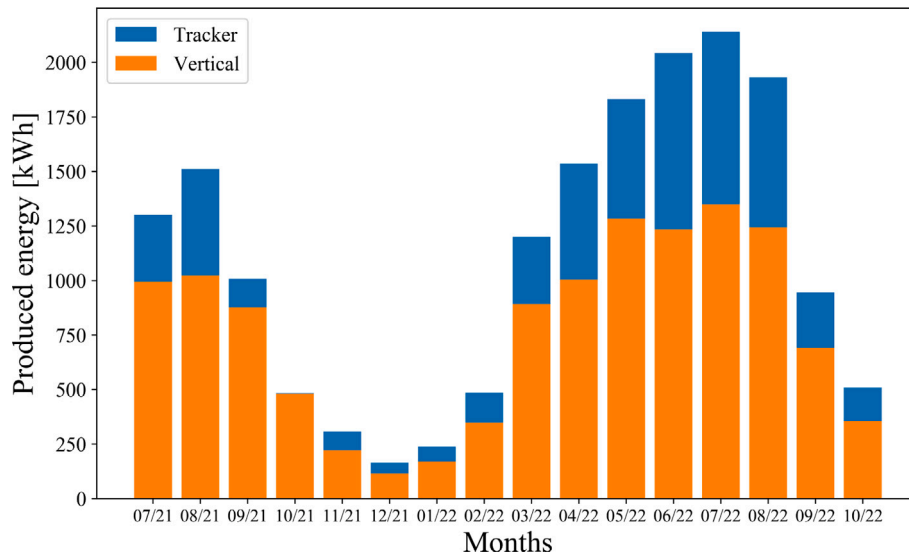


Fig. 12. Monthly AC electricity production of the Grembergen pilot project, showing the advantage (surplus electricity) of the solar tracker compared to the vertical set-up while the same PV capacity (450 kWp/ha) is installed. In 2021, the smart-tracking is applied and in 2022 the standard solar tracking, visible by the additional gain compared to the vertical set-up.

and with lower crop yield losses, the LER for the vertical fixed setup increased by 0.26 points (reflecting a 26% increase in relative crop yield compared to 2021). Similarly, the LER for the tracking setup increased by 0.32 points (indicating a 20% increase in electricity yield due to full tracking and a 12% increase in relative crop yield compared to 2021). These findings demonstrate (and approve) that agrivoltaics holds greater potential in dry and hot regions.

3.4. Financial performance and business model

For the investor, which can be the farmer or another third-party, energy market prices showed to be highly volatile, increasing enormously the risk and return on investment. Therefore, as suggested by other work [22], a fixed purchase power agreement (PPA) price between a large off-taker and the owner of the agrivoltaic site is highly recommended and desirable business model for agrivoltaic systems.

Table 1

LER values interspersed agrivoltaic installations in Grembergen for 2021 and 2022 given a land loss of 22%.

	2021	2022
relative crop yield vertical (%)	74	100
relative energy yield vertical (%)	40	40
land loss vertical (%)	22	22
LER vertical	0.95	1.21
relative crop yield tracking (%)	78	90
relative energy yield tracking (%)	55	75
land loss vertical (%)	22	22
LER tracking	1.15	1.47

The Levelized Cost of Electricity (LCOE) plays a pivotal role in a PPA model due to its significance in determining the financial feasibility and attractiveness of renewable energy projects as the LCOE is a metric that

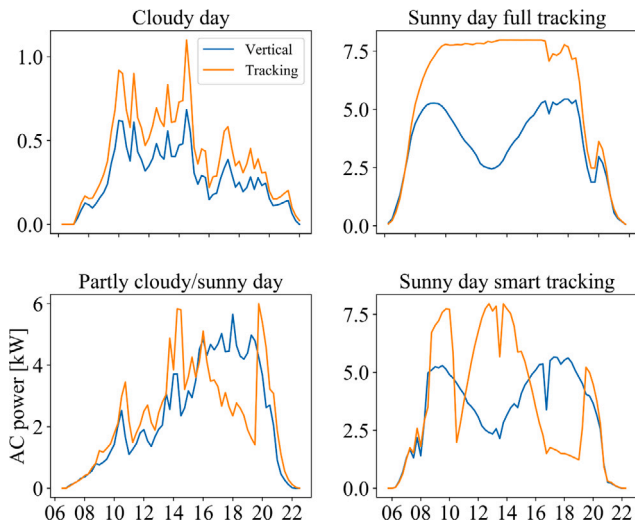


Fig. 13. Daily production profiles (power in Watt) for different sky conditions. The blue line is the vertical bifacial set-up and the orange line the solar tracker. The drops in power at 10 am and 18 pm indicate the anti-tracking moments in the smart tracking algorithm.

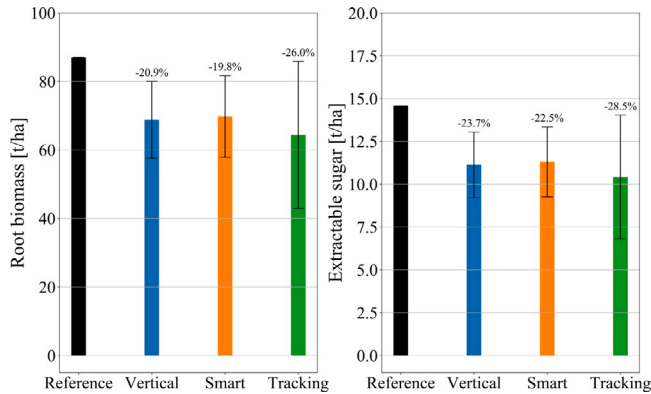


Fig. 14. Simulated sugar beet yield and quality in agrivoltaic and full light conditions without consideration of the land loss based on an empirical sugarbeets model. The bar graphs show that in perfect watering conditions (no water stress), shade of PV modules results in a loss of biomass.



Fig. 15. Picture during cultivation of the field in Grembergen, showing the land loss in agricultural land due to the presence of the agrivoltaic system (construction + border of 1 m at each side where the machinery is not able to cultivate to avoid collision with the structure).

Table 2

Costs interspersed agrivoltaic installations in Grembergen.

	Vertical set-up	Smart tracking	Full tracking
CAPEX	€1/Wp	€1/Wp	€1/Wp
OPEX	1.5% [25]	2.5% [26]	2.5% [26]
LCOE	€100/MWh	€77/MWh	€73/MWh
ΔOPEX _{agri}	+ €30	+ €30	+ €30

calculates the per-unit cost of producing electricity over the lifetime of the project, taking into account all upfront capital costs, ongoing operational expenses, maintenance, and financing.

The total cost of the agrivoltaic installations can be estimated based on two variables: the levelized cost of energy for the PV system (LCOE) and the levelized cost of energy for the Agri-PV system (LCOE_{Agri}), which add the agricultural costs to the traditional LCOE. The capital expenditure (CAPEX) for the vertical bifacial and single-axis tracking setups is determined by the costs incurred during their construction and is assumed to be equal (as shown in Table 2). However, the tracker system requires additional expenses for the programmable logic controller (PLC) system and actuators. It should be noted that the tracker system requires fewer vertical piles compared to the fixed vertical setup, as it can go flat when wind speeds exceed 15 m/s to minimize wind load. On the other hand, the operational expenditure (OPEX) for the tracker system is considered higher due to the presence of moving parts, associated maintenance, and increased need for cleaning to mitigate soiling effects compared to the vertical bifacial modules [23].

To calculate the LCOE as shown in Equation (2), annual cash flows are discounted using a weighted average cost of capital (WACC) of 6% [24]. Furthermore, a degradation rate (d) of 1% is incorporated to account for the aging of the PV modules. The LCOE is calculated over a time period (N) of 30 years for glass–glass modules. The LCOE for the vertical setup is estimated to be €100/MWh, while for the tracker system, it ranges between €73–77/MWh depending on the specific years considered and the choice between smart or full solar tracking.

$$LCOE_{agriPV} = \frac{CAPEX + \sum_{t=1}^N \frac{OPEX}{(1+WACC)^t}}{\sum_{t=1}^N \frac{Y_{el,AV} \times (1-d)^t}{(1+WACC)^t}} + \frac{\sum_{t=1}^N \frac{(Y_{agri,ref} - Y_{agri,AV} \times (1-L)) \times S_{agri} + OPEX_{agri}}{(1+r)^t}}{\sum_{t=1}^N \frac{Y_{el,AV} \times (1-d)^t}{(1+r)^t}} \quad (2)$$

In terms of LCOE_{Agri} based on the data in 2021, the fixed vertical and smart tracking setups experienced a reduction in available land (land loss) and diminished light availability for the sugar beets. This led to a decrease in crop yield of 30 tons/ha and 26 tons/ha, respectively. Considering an average selling price of €40/ton, this translates to a loss of €1400/ha and €1200/ha per year for the fixed vertical and smart tracking setups, respectively. Additionally, the presence of the PV structure increased farming costs by €30/ha per year, accounting for approximately 5% of the average farming expenses per hectare. Consequently, this resulted in an additional operational expenditure (OPEX) of €3.1/kWp and €2.7/kWp, and an increase in the “traditional” LCOE by €3.5/MWh and €2/MWh for the fixed vertical and smart tracking setups, respectively.

Regarding the economic performance in the year 2022, the reduction in sugar beets compared to the reference field was minimal due to water stress, resulting in a negligible impact (almost no increase) on the traditional LCOE.

3.5. Practical challenges during the exploitation

During the construction of agrivoltaic systems, practical challenges related to cable depth and soil compaction need to be considered. Firstly, cable depth is an important factor to address. Cables for electrical connections in the system must be buried at a sufficient depth



Fig. 16. Pictures showing the additional operating expenditures in agrivoltaic systems: bird droppings on the modules on the left and middle pictures and weed growth on the uncultivated area causing module shading on the right picture.

to prevent damage during plowing or other agricultural activities. The depth should be carefully determined to ensure that the cables remain protected and do not interfere with farming operations. Proper planning and coordination with farmers and installation teams are necessary to determine the appropriate depth and avoid potential issues. Secondly, soil compaction is a common challenge during the construction phase. Heavy machinery used for installing the PV structures and other components can cause soil compaction, leading to adverse effects on soil health and crop growth. Compacted soil restricts root growth, reduces water infiltration, and limits nutrient availability. It can result in poor crop development and yield reduction.

When it comes to the challenges during the exploitation of interspersed agrivoltaic systems, it was observed that both systems were susceptible to bird droppings on multiple occasions throughout the season (Fig. 16, left and middle pictures). These bird droppings can lead to increased mismatch losses and module damage, including the formation of hotspots. Additionally, weed growth near the structures (Fig. 16, right picture) necessitates additional mowing every six months to prevent shading of the PV modules.

These observations indicate that agrivoltaic installations may incur higher operating expenses (OPEX) compared to other PV applications or traditional agriculture without PV modules, particularly in the context of industrial farming. The presence of bird droppings and weed growth poses unique challenges in these settings, where larger areas are being cultivated and managed intensively. Such challenges may require additional measures to mitigate their effects, such as the installation of spikes on top of the vertical set-up or the use of special low-height species to deter birds and reduce droppings. These measures are crucial to minimize the impact of bird droppings and weed growth, ensuring optimal performance of the agrivoltaic system. It is essential to consider these challenges and implement appropriate strategies to maintain system efficiency and manage the associated OPEX costs, especially during periods of drought or limited rainfall events [27].

4. Discussion

In this section, the discussion regarding the performance difference and investment decision, the model's performance, the measured data, and the employed setup algorithm is given. It highlights the system's strengths and limitations, facilitates the refinement of models and algorithms, and guides future research and practical implementations in the field of agrivoltaics.

4.1. What are the performance differences between fixed vertical bifacial setups and dynamic single-axis systems in interspersed agrivoltaics and what is the impact on the investment decision and business model?

Despite the higher OPEX costs, the single-axis bifacial solar tracker with full tracking and smart-tracking algorithm results in the lowest levelized cost of electricity for similar crop yields compared to the vertical fixed set-up (mainly due to the increase in the specific

electricity production) [28]. This means that solar tracking technology needs the lowest subsidy and or off-take strike PPA price to be financially attractive. The estimated agrivoltaics LCOE prices lay in line with reported European PPA prices of PV projects, however are higher than reported ground-mounted PV PPA prices in the Netherlands between 56-60€/MWh (<https://www.pv-magazine.com/2023/06/13/ppa-prices-falling-across-europe-says-pexapark/>), meaning there is an agrivoltaic price premium on the off-take price that needs to be considered.

However, besides the value of the PPA price, also the risk of uncertainties is important. One disadvantage of the dynamic agrivoltaic system compared to the static vertical one, is its probability of (mechanical) failures during operation reducing the expected energy generation. Also, the flexible “smart tracking” algorithm during the growing season highly depends on the type of crop, creating additional uncertainty/variability in the expected generation, costs and (agricultural) revenues during its lifetime, which impacts drastically the risk of the investment.

Indeed, the solar tracker is able to adapt the light conditions to specific needs (i.e. homogeneity, time of growing season, crop type), which is advantageous with regard to a crop rotation system, common in arable farming. Also, with regard to the Belgian (variable) weather patterns (alternating dry and wet seasons), the tracker provides the necessary flexibility.

4.2. What are the discrepancies between the crop model and actual field measurements?

The empirical crop model that includes shadow response curves show the relationship between the percentage of shading and the corresponding decrease in crop yield. However, this study shows that shadow response curves are species-specific and may not accurately reflect the actual growth and yield of the crops in the agrivoltaic system. As we had 2 different seasons, other factors such as irrigation, temperature, and soil fertility showed to have a significant impact on crop growth and yield, and should be taken into consideration when predicting crop yield in agrivoltaic systems. As a result, shadow response curves should be used in combination with other methods and data, such as field measurements and simulations, to provide a more complete and accurate picture of crop yield in agrivoltaic systems.

4.3. How do we need to interpret the measured data?

The two-year crop data collected for the study exhibited significant influences from soil compaction during the construction phase and variations in weed availability, as well as differences in weather patterns between the two years. These factors can have notable implications for the interpretation and generalizability of the results: soil compaction during construction can impact crop growth and yield by affecting soil structure, nutrient availability, and root development while weed availability and competition can also play a crucial role in crop performance.

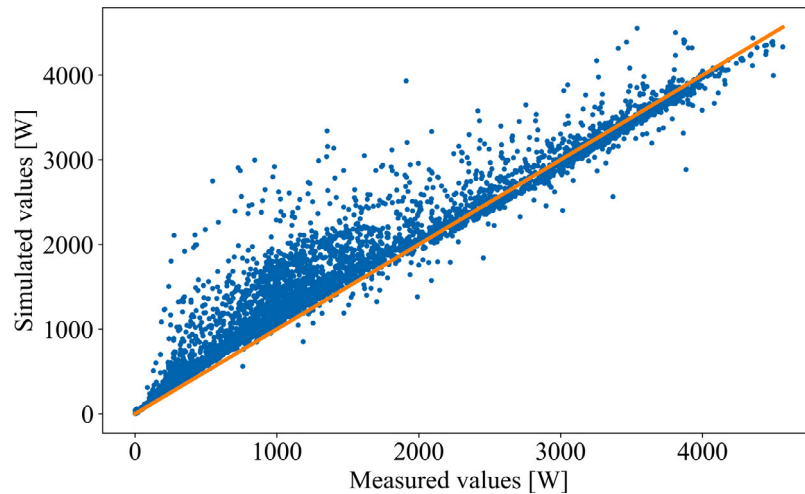


Fig. 17. Comparison between simulated and measured DC power values of the vertical setup in Grembergen.



Fig. 18. Visible structure shading by the upper mechanical profile on the PV module causing a mismatch between simulated (unshaded) and measured (shaded) power values.

The real-time power measurements are evaluated based on a simplified PV model that includes the measured irradiance and module temperature in Fig. 17. This picture depicts a scatter plot that reveals an overestimation of power production by the simulation model for measured power values below 2000 W.

Upon conducting a thorough analysis of the deviation between the expected and measured yield, it became evident that this discrepancy could be attributed to shade caused by the structures on the backside of

the PV modules. This shading leads to mismatch losses and a reduction in power output.

To mitigate this issue, POA reference cells were positioned on top of the mounting structure, ensuring that they remained unaffected by structure shade. Fig. 18 provides a visualization of the structures causing shading on the backside of the PV modules. In order to improve the accuracy of energy yield predictions, an hourly varying structure shading factor was incorporated into the model for both the vertical and tracking setups. This shading factor was calibrated using measured data, employing a training set (comprising 75% of the data) and a test set (comprising 25% of the data), with the objective of achieving the highest possible R^2 value. This approach aligns with the methodology proposed by Zhao et al. [29].

In the tracking system with a DC/AC ratio of 1.36, the AC measurements reveal yearly clipping losses of 2% when calculated using hourly TMY-data. In contrast, the vertical bifacial system with a DC/AC ratio of 1.82 experiences clipping losses of 3.5%. It can be concluded that careful attention should be given to the sizing of PV inverters, which significantly varies depending on the system type. Nonetheless, the trade-off between clipping losses and a higher power rating for inverters is primarily a financial consideration rather than a technical one.

For the electrical data, using the calibrated energy prediction model, which incorporates structure shading factors, the specific yield was found to be 10% lower for the vertical system and 7% lower for the tracker setup compared to theoretical expectations. This observation highlights the greater significance of using frameless modules in a vertical setup, as suggested by Rabanal et al. [30]. Specifically, in the morning (or afternoon, depending on the installation), the back of the bifacial module receives direct light, resulting in shading from the module's frame. In a tracking setup, the back of the PV module is predominantly exposed to reflected and diffuse light, reducing the impact of shading. These significant findings emphasize the substantial influence of structure shading on system performance.

Finally, the LCOE calculation requires the consideration of project costs and the total energy generated over the project's lifetime. However, using only two years of data may not provide a comprehensive representation of the system's performance throughout its entire lifespan. The short-term nature of the data may not capture long-term trends or account for exceptional weather events (and impact on crop financial losses or gain) that can occur over several years. In the context of a standard solar PV business model centered around a (corporate) power purchase agreement (PPA), where the generated electricity is sold over the grid to a large consumer, the incorporation of annual production variability finds its expression through P50 and P90 metrics.



Fig. 19. East-west single-axis solar tracker in different positions at the 13th of July, around 12 h 30 pm. The pictures show that at that moment of the day, the shade falls below the tracker, unable to protect the crops in between against the high amount of solar radiation.

However, in the case of an agrivoltaic PPA initiative, it becomes imperative to encompass the fluctuation in agricultural revenues and costs as well, a critical step aimed at fostering investor interest. The unpredictability in agricultural yield and associated costs, coupled with the inherent challenges of implementing agrivoltaic systems, exert a notable impact on the variability of outcomes and, consequently, the determination of purchase power agreement strike prices. These factors necessitate a comprehensive assessment that considers not only the energy generation variability but also the intricate interplay between agricultural and financial aspects to arrive at a balanced and realistic pricing strategy.

4.4. Improvement agrivoltaic set-up and tracking strategies

The smart tracking algorithms evaluated in this study optimize the daily photosynthetically active radiation (PAR) and its spatial distribution, minimizing the need for anti-tracking periods. This calendar-based approach simplifies the tracking strategy without relying on complex crop growth models integrated into the system. By using this approach, the system can adapt to different crop types and growing seasons, reducing uncertainties in annual revenues and energy generation over the system's lifetime. Another tracker strategy suggested in the literature is periodic anti-tracking, which considers specific crop development stages that are critical for quality or yield [31]. However, this approach introduces additional uncertainties due to the variability in crop stage duration, which depends on cumulative air temperature [32]. The study also demonstrates that low-density interspersed agrivoltaic systems already provide a relatively high amount of field PAR under full tracking mode.

As shown in Fig. 19, one limitation of low-density interspersed systems is their limited capacity to protect against hot and dry periods. Due to their east-west configuration, low positioning, and large pitch, these systems cannot provide shade to the entire cultivation area during midday and evening. In semi-arid areas, a denser overhead agrivoltaic system, as proposed by Valle et al. [13], would be more suitable as it can cover and protect a larger portion of the cultivated area. Additionally, overhead systems offer better control over microclimatic factors such as rain/hail distribution, wind profile, and air temperature and humidity. However, the increased capital expenditure (CAPEX) associated with overhead systems, primarily due to the need for additional steel, significantly raises the levelized cost of electricity (LCOE) and reduces its financial competitiveness compared to ground-mounted PV systems.

5. Conclusion

In this study, we assessed the technical feasibility of a dynamic single-axis and fixed vertical bifacial interspersed agrivoltaic system.

Design parameters such as row distance and facility orientation were fixed based on practical considerations such as machinery dimensions and field layout. After optimization using ray-tracing software to increase the bifacial gain, two set-ups with equal power density in a double landscape position were designed and constructed.

The results of this study demonstrate that interspersed agrivoltaic systems, have a reasonable business case in terms of levelized cost of energy (LCOE) compared to ground-mounted PV projects, mainly attributed to their lower CAPEX such as steel requirements in overhead agrivoltaics. However, it is important to note a significant limitation compared to overhead agrivoltaic systems, as the low-ground and large pitch design of interspersed systems is unable to provide complete protection for the entire cultivation area against extreme conditions.

From both technical and economic perspectives, the single-axis tracker proves to be a preferable option over the vertical setup. In addition to higher annual energy production and lower LCOE, the tracking setup offers flexibility in increasing cumulative photosynthetically active radiation (PAR) and optimizing its distribution across the field through the use of anti-tracking modes tailored to specific growing seasons and crop types. A novel diurnal smart-tracking approach was implemented to maximize daily PAR exposure while minimizing the need for anti-tracking periods, resulting in optimized light use efficiency (LER) under optimal watering conditions.

In this study, we employed an empirical shade response curve based on sugar beet cultivation under optimal water and pest-free conditions, which resulted in a reduction in sugar beet yield. Additionally, significant land loss around the agrivoltaic structures was observed, preventing the farmer from cultivating the area and resulting in an LER value lower than one and no increase compared to separated production systems. However, during the dry and hot season of 2022, the same shade levels resulted in minimal crop yield losses, leading to an improvement in LER by 20%. These findings suggest that the empirical shade response curve varies depending on the climatic conditions of each year.

Furthermore, our study highlights the practical challenges associated with combining PV and agricultural activities using current conventional farming techniques with large machinery. Attention must be given to the placement of cables at greater depth compared to ground-mounted PV systems. Additionally, during operation, significant land losses occur as farmers are cautious about potential collisions with the structures. The uncultivated land area results in crop losses and increased weed pressure, which shades the PV modules and requires additional maintenance. Furthermore, bird droppings were observed, necessitating additional cleaning. These practical considerations as well as the variability in agriculture yield and associated costs demonstrate that the integration of PV and agricultural activities in current practices is not as straightforward as initially envisioned and not as easy as PV combined with grazing (rangevoltaics) or biodiversity (ecovoltaics).

Overall, this study provides valuable insights into the technical and economic aspects of interspersed agrivoltaic systems, highlighting the advantages of single-axis tracking and the challenges associated with practical implementation. Further research should focus on addressing these challenges and developing innovative solutions to maximize the benefits of agrivoltaics by a synergetic crop choice and climate while minimizing the practical limitations encountered during operation and the risk for the investor in agricultural yield variability.

CRedit authorship contribution statement

Brecht Willockx: Methodology, Software simulation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Cas Lavaert:** Methodology, Software simulation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Jan Cappelle:** Formal analysis, Supervision.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Brecht Willockx reports was provided by KU Leuven.

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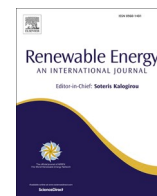
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Corrigendum to “Performance evaluation of vertical bifacial and single-axis tracked agrivoltaic systems on arable land” [Renew. Energy 217 (2023) 119181]

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The authors regret the inaccuracies in the published version of the paper, notably, the following points were not correctly presented.

1. The authors acknowledge inaccuracies in the numerical data presented in section 3.1 and propose the replacement of this section with the following text:

“Fig. 12 presents the measured monthly AC electricity production (kWh) for each system, both operating at equal power density. The data clearly demonstrates that the dynamic single-axis tracker outperforms the fixed vertical setup, resulting in approximately a 35% increase in monthly electricity production. Notably, the difference in tracking strategy between 2021 and 2022 is evident. For instance, comparing the months of July, August, and September in 2021 (smart tracking) to the same period in 2022 (full tracking), the electricity production shows a significant disparity, with a +15% increase in 2021 and a more substantial +35% increase in 2022.

Over the course of the first year of operation, from July 2021 to July 2022, the specific yield for the dynamic single-axis tracker was 1305 kWh/kWp, while the fixed vertical bifacial setup achieved a specific yield of 915 kWh/kWp. These specific yield values provide insights into the energy production efficiency of each system, highlighting the superior performance of the dynamic single-axis tracker. As a reference typical South-facing ground mounted installation, with the same bifacial modules, a tilt of 20° and ground coverage ratio of 55%, the specific yield would be 1045 kWh/kWp. However, given the low ground coverage ratio of the agrivoltaic installations (resulting in 450 kWp/ha) compared to the reference South-facing installation (1140 kWp/ha), the electricity yield per ha is 65% lower for the vertical bifacial system and 50% lower for the single-axis tracker, both compared to the reference South-facing installation.

Fig. 13 illustrates the daily production profiles under different sky conditions. An initial observation is the similar start and end times of production for both the dynamic single-axis tracker and the fixed vertical setup, despite the mechanical limit of the tracker being 50° and the vertical setup being fixed at a 90° angle. On cloudy days, the tracker is capable of assuming a flat

position, enhancing light interception and electrical production by approximately 50% compared to the vertical setup. In sunny conditions, the dynamic tracking system, which continuously optimizes the angle of incidence of sun rays, outperforms the vertical setup, particularly during midday when the difference is most pronounced. Fig. 13 also highlights the disparity between full energy tracking and smart tracking. During anti-tracking moments (indicated by the drops at 10 a.m. and 18 p.m.), the electrical production diminishes to approximately 25% of the nominal power.”

2. The authors regret that the numerical data presented in Table 1 is inaccurate, and they recommend replacing this table with the following:

	2021	2022
Relative crop yield vertical (%)	74	100
Relative energy yield vertical (%)	35	35
Land loss vertical (%)	22	22
LER vertical	0.93	1.13
Relative crop yield tracking (%)	78	90
Relative energy yield tracking (%)	46	54
Land loss vertical (%)	22	22
LER tracking	1.07	1.24

3. In light of the inaccuracies found in the data within Table 1, it is necessary to revise the accompanying text describing this table. The authors propose amending section 3.3 with the following text:

“The dual land use of agrivoltaics can increase the spatial efficiency. A commonly used parameter in agrivoltaics to express the spatial efficiency is the Land Equivalent Ratio (LER) which can be calculated according to Equation (1).

The LER compares the electrical and agricultural yields ($Y_{el,AV}$ (MWh) and $Y_{agri,AV}(kg)$) per unit of land (ha) between agrivoltaic and reference conditions. Consequently, the values of the LER are highly dependent on the

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reference situation. In this study, the reference ground-mounted PV is considered with the same type of bifacial modules, where the PV modules do not fully cover the agricultural area. The coverage ratio (55%), height (1m) and tilt (20°) are adapted to have at least 30% available sunlight, maintaining soil fertility and biodiversity. This adaptation is needed since the LER does not evaluate the environmental impact between a reference and agrivoltaics solar farm. The same reasoning is made for the agricultural yield. It is the marketable yield (kg/ha) that is compared with the reference (i.e. quality parameters such as sugar content, shape, color are included). Important is to reduce the agricultural area in agrivoltaic conditions by a landloss factor LL (between 0 and 1) due to the presence of the agrivoltaic structure and borders.

Theoretical calculations initially accounted for a buffer zone of 0.5 m on both sides of the agrivoltaic structure, which would result in a land loss factor of 11% (1 m divided by 9 m row distance). However, as depicted in Fig. 15, in practice, the buffer zone was measured as 1 m on each side due to the absence of GPS-tracked machinery. Consequently, the practical land loss amounted to 22%.

As tabulated in Table 1, based on the significant land loss, reduction in root biomass and reduction in electricity yield per ha, the vertical system in the rain-fed 2021 season achieved a Land Equivalent Ratio (LER) of 0.93. This indicates that there was no improvement in spatial efficiency compared to separate production systems. In contrast, the smart tracking system achieved an LER of 1.07, primarily due to its higher specific energy yield when compared to the vertical system.

In the dry 2022 season, which allowed for full tracking capability resulting in a relative energy yield of 54% compared to the reference system, and with lower crop yield losses, the LER for the tracking setup increased by 0.17 points. Similarly, the LER for the vertical fixed setup increased by 0.2 points (reflecting a 26% increase in relative crop yield compared to 2021). These findings demonstrate (and approve) that agrivoltaics holds greater potential in dry and hot regions.”

4. The authors recommend revising the highlights based on the data in Table 1 to:

“Highest increase in land efficiency (24%) measured in hot, dry conditions.”

5. The abstract should also be revised to replace the current representation of results with the following text:

“This study evaluates and compares two types of interspersed bifacial agrivoltaic systems in Belgium: a fixed vertical system and a dynamic single-axis tracker, focusing on sugarbeet cultivation. Additionally, the impact of different tracking algorithms on crop yield and quality is investigated. The main approach involves developing an empirical crop model based on radiation use efficiency (RUE) and utilizing a radiation tool to simulate crop yield and quality. Field measurements of electricity and crop output are conducted over a span of two years, 2021 and 2022. The findings reveal that the dynamic solar tracker outperforms the fixed vertical bifacial setup in both years. The smart-tracking algorithm, applied under optimal watering conditions in the 2021 season, leads to a significant increase in energy yield (+30%) and land use efficiency (+15%) at a lower cost, while maintaining comparable crop yields to the vertical setup. However, limitations are observed in the empirical crop growth model’s ability to account for climatic variability in the dry 2022 year, reducing its usefulness during the design phase. Notably, the practical implementation of the systems highlights challenges during construction and exploitation, emphasizing the significance of considering these practical factors together with climate, structure and crop choice assessing the effectiveness and to de-risk investments.”

6. A statement should be added to acknowledge that Brecht Willockx and Cas Lavaert share first authorship.

The authors would like to apologise for any inconvenience caused.